Xi'an Jiaotong-Liverpool University

西交利物浦大学

MODULE CODE	EXAMINER	DEPARTMENT	TEL
CPT101	STEVEN GUAN	COMPUTING	1501

1st SEMESTER 2023/24 Open-Book FINAL EXAMINATIONS

BACHELOR DEGREE – Year 2

COMPUTER SYSTEMS

TIME ALLOWED: 2 Hours

INSTRUCTIONS TO CANDIDATES

- **This is an open-book exam.**
- **Total marks available are 100.**
- **Answer all questions.**
- **No calculator is allowed during the examination.**
- **Only answers in English are accepted.**

THIS PAPER MUST NOT BE REMOVED FROM THE EXAM HALL.

Part I. (60 marks)

Single-Choice Questions (40 marks)

1. Which of the following best describes the Input-Process-Output model?
A. A physical representation of hardware structure
B. A conceptual model showing how computers receive, process, and output data
C. A programming model used for compilers

2. In the von Neumann model, both program and data are stored in:
A. Separate memories
B. CPU registers
C. The same memory space

3. The main difference between von Neumann and Harvard architectures is that Harvard architecture:
A. Uses different memories for programs and data
B. Stores data and instructions in the same place
C. Is slower than von Neumann

4. Which of the following components performs actual calculations and operations?
A. Memory
B. CPU
C. Hard disk

5. Which memory type is volatile and holds data temporarily?
A. RAM
B. Hard disk
C. ROM

6. Which device is not an input device?
A. Mouse
B. Monitor
C. Scanner

7. The von Neumann bottleneck refers to:
A. Slow CPU speed
B. Limited transfer rate between memory and CPU
C. Software translation delay

8. High-Level Languages (HLLs) must be translated into machine code because:
A. CPUs can only execute binary instructions
B. HLLs are faster
C. HLLs are hardware-dependent

9. Which of the following is not a category of machine instruction?

A. Input-Output
B. Data transfer
C. Data encryption

10. Who is credited with formulating the concept of the stored-program computer?
A. John von Neumann
B. Alan Turing
C. Konrad Zuse

11 In information theory, Shannon introduced the concept of entropy to describe:
A. The randomness and uncertainty of information.
B. The physical energy needed to transmit signals.
C. The accuracy of data representation in memory.

12 Which of the following best defines the relationship among data, information, and knowledge?
A. Knowledge organizes data into useful form.
B. Information is analyzed data; knowledge is raw data.
C. Data are raw facts; information organizes them; knowledge applies reasoning.

13 The binary sequence 10110010011_2 represents the decimal number:
A. 1427 B. 1023 C. 2397

14 When converting from binary to decimal, the process primarily involves:
A. Repeated division by 2.
B. Multiplying each bit by its positional weight and summing.
C. Bitwise negation and accumulation.

15 The hexadecimal number $2AE_{16}$ equals:
A. 670_{10} B. 686_{10} C. 720_{10}

16 Which statement about ASCII is TRUE?
A. It uses 7 bits and defines 256 unique characters.
B. It uses 8 bits and includes international symbols by default.
C. It uses 7 bits and is extended to 8 bits for additional characters.

17 Which of the following is a limitation of ASCII?
A. It cannot represent all characters of global languages.
B. It is slower than EBCDIC in transmission.
C. It requires parity bits for every letter.

18 The Unicode standard allows encoding of up to approximately:
A. 256 characters B. 65,536 characters C. 1 million characters

19 The Two's Complement representation is used mainly because:
A. It simplifies subtraction and negative number handling.
B. It doubles storage efficiency.
C. It ensures floating-point precision.

20 According to the IEEE-754 standard, floating-point numbers consist of:
A. Sign, mantissa, exponent.
B. Sign, integer, fraction.
C. Base, exponent, checksum.

Multiple-Choice Questions (20 marks)

Which statements correctly describe ASCII control characters?
A. 'BEL' triggers a bell sound.
B. 'HT' controls horizontal tab position.
C. 'SOH' indicates end of transmission.
D. 'NUL' represents a zero character.
E. They occupy code values below 32 decimal.

Regarding status flags (EFLAGS register) in x86:
A. Z = 1 indicates zero result.
B. S = 1 indicates negative result.
C. O = 1 indicates signed overflow.
D. C = 1 indicates carry or unsigned overflow.
E. All are cleared by MOV instruction.

In assembly, the stack (ESP) behaves as:
A. A LIFO structure.
B. PUSH decreases ESP value.
C. POP increases ESP value.
D. It is used for subroutine calls and parameter passing.
E. Items pushed are removed in FIFO order.

Correct statements about assembly instruction syntax include:
A. Labels must end with a colon.
B. Comments start with a semicolon (;).
C. Every instruction must have at least one operand.
D. Mnemonics specify the operation to perform.
E. A program is formed by multiple instructions executed sequentially.

Part II. (40 marks)

25. Fill-in-the-blank question (10 marks)

In the von Neumann model, both _____ and _____ are stored in the same memory space.

Moore's Law states that the number of _____ on a chip roughly _____ every _____ years, leading to faster and smaller _____.

A library file contains _____ code that can be linked into a program.
Static libraries are linked _____, while dynamic libraries are linked _____.
Dynamic linking avoids _____ of library code in memory.

26. Drag-and-drop (for online test) or write the sequence number (for on-site test) of the assembly code to form a program that finds the **sum of odd** numbers between **1 and 100** and stores this value in the eax register. Note that your sequence must absolutely match the line numbers to the left-most column of the table. The answers for Lines 2,3, and 5 have been provided. Complete the rest. **(30 marks)**

	Correct Sequence	**Pick From Here**	
Line 1		1	mov eax, 0
Line 2	2	2	mov ebx, 1
Line 3	3	3	mov ecx, 100
Line 4		4	mov eax, 1
Line 5	5	5	jz L2
Line 6		6	L1: and ebx, 00H
Line 7		7	L1: and ebx, 01H
Line 8		8	L2 : inc eax
		9	L2 : inc ebx
		10	L1: test ebx, 00H
		11	L1: test ebx, 01H
		12	loop L1
		13	add eax, ebx
		14	loop L2

END OF PAPER